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Unbiased Tandem System for Simultaneous Methane and Glycerol Upgrading to Formic Acid Under Ambient Conditions

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Received: 29 May 2026 | **Revised:** 25 August 2026 | **Accepted:** 15 September 2026

Keywords: methane | in situ chemical oxidation | radical

ABSTRACT

Direct methane conversion to liquid products under mild conditions remains a long-standing challenge, often requiring expensive oxidants or external bias that hinder practical viability. Here, we report a bias-free photovoltaic-photoelectrochemical (PV-PEC) tandem system that co-upgrades methane and glycerol into formic acid at both electrodes under ambient conditions. The cathodic process employs oxidized carbon black to drive in situ generation of hydrogen peroxide, which in turn mediates methane oxidation via electro-Fenton reactions in the presence of Fe^{2+}. The photoanodic process uses nanoporous $BiVO_4$ to oxidize glycerol with nearly 100% Faradaic efficiency toward a mixture of glycerol oxygenates. Coupling these reactions in a PV-PEC configuration enables bias-free operation at 1.7 mA cm^{-2} under standard solar illumination (Air Mass 1.5 Global, AM1.5G), yielding 2.87 mmol g_{cat}^{-1} h^{-1} formic acid from methane oxidation (with a total production rate of 7.99 μmol cm^{-2} h^{-1}) and achieving Faradaic efficiencies of 4.96% for the cathode and 28.5% for the anode. Life cycle assessment reveals over 66% reduction in carbon footprint compared to conventional syngas-based processes. This work demonstrates the first bias-free, full-cell platform for simultaneous methane and glycerol upgrading, advancing sustainable chemical production and efficient biomass utilization.

1 | Introduction

Methane (CH_4) is the principal component of natural gas and biogas, representing both a valuable carbon feedstock and a potent greenhouse gas with a global warming potential 28 times that of CO_2 over 100 years [1, 2]. While CH_4 is primarily combusted for heat and power generation, its selective transformation into liquid fuels or chemicals offers an attractive strategy for decentralized carbon utilization [3, 4]. Typically, industrial CH_4 conversion relies on high-temperature, high-pressure steam reforming to produce synthesis gas (syngas), followed by multiple catalytic steps to reach the target molecule [5–7]. Such multi-step pathways are energy-intensive, require large-scale complex infrastructure, and release substantial CO_2 emissions.

Efforts to enable direct CH_4 conversion under mild conditions have included heterogeneous catalysis, photochemical methods, and electrochemical approaches [8–10]. Among them, photocatalytic CH_4 oxidation can achieve high selectivity toward partial oxidation products, but often requires external oxidants such as hydrogen peroxide (H_2O_2), which are costly and unsustainable for large-scale use [11–15]. Electrochemical approaches offer the possibility of in situ oxidant generation, such as H_2O_2 from the two-electron oxygen reduction reaction ($2e^-$ ORR), followed by

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hydroxyl (•OH) and hydroperoxyl (•OOH) radical formation in the electro-Fenton process for CH_4 activation [16–18]. However, most systems reported previously operate in half-cell configurations with the oxygen evolution reaction (OER) as the counter process, which is kinetically sluggish and requires substantial overpotential, lowering overall energy efficiency.

Replacing the OER with a more favorable anodic reaction can reduce the cell voltage and simultaneously yield co-products of value [19]. Biomass-derived molecules, such as glycerol, a low-cost byproduct of biodiesel production, are particularly attractive due to their lower oxidation potential [20] and ability to yield oxygenated chemicals [21, 22]. In prior work, improved economics and life cycle were demonstrated for electroreduction of CO_2, when replacing OER with the glycerol oxidation reaction (GOR) [19]. The GOR can proceed at potentials hundreds of millivolts lower than OER, significantly reducing energy intensity. Furthermore, GOR can produce compounds including formic acid (HCOOH), glyceric acid, glycolic acid, and dihydroxyacetone, all value-added chemicals that have established industrial use [23–26].

Here, we report the first bias-free tandem device that couples the methane oxidation reaction (MOR) at the cathode with GOR at the photoanode, producing HCOOH as the major product at both electrodes under ambient conditions. The cathodic process undergoes the $2e^-$ ORR to generate H_2O_2 in situ, which in turn reacts with Fe^{2+} via the electron-Fenton process to produce •OH radicals for CH_4 activation. The photoanodic process oxidizes glycerol under illumination, providing the anodic current to drive the cathodic reaction without external bias. This synergistic pairing eliminates the need for external oxidants, avoids the high overpotential of the OER, and enables efficient co-production of high-value liquids derived from both CH_4 and glycerol. Using a photovoltaic-photoelectrochemical (PV-PEC) architecture, the system achieves a CH_4-derived HCOOH production rate of 2.87 mmol g_{cat}^{-1} h^{-1} for the cathode side and a total HCOOH rate of 7.99 μmol cm^{-2} h^{-1} for the full cell. Life cycle assessment (LCA) suggests a carbon footprint more than 66% lower than that of the conventional syngas route, offering a promising pathway for decentralized chemical manufacturing from abundant carbon and biomass resources.

2 | Results and Discussion

2.1 | Integrated MOR-GOR Device Concept

The MOR-GOR full-cell system integrates two complementary reactions within a single device to directly convert CH_4 and glycerol into value-added liquid products under ambient conditions. Unlike the conventional multi-step syngas route (Figure 1a), which requires high-temperature reforming and multiple energy-intensive thermocatalytic steps, our approach performs both CH_4 and glycerol conversion in a unified, room-temperature process (Figure 1b). In the cathodic compartment, a co-feed of CH_4 and O_2 drives the $2e^-$ ORR to electrochemically produce H_2O_2. Dissolved Fe^{2+} then catalyzes the Fenton reaction, generating •OH radicals that activate CH_4 to methyl radicals (CH_3•), which are subsequently oxidized to oxygenates such as HCOOH. In the photoanodic compartment, glycerol undergoes selective oxidation under illumination, generating HCOOH and other oxygenates, while supplying the anodic current that drives cathodic H_2O_2-mediated CH_4 oxidation. This dual-reaction design eliminates the need for external oxidants, bypasses the high overpotentials of OER, and maximizes overall carbon efficiency by simultaneously producing high-value liquids at both electrodes.

From a thermodynamic standpoint, the feasibility of this coupled system can be rationalized by considering the relative redox potentials of the anodic and cathodic processes (Table 1). On the cathodic side, the $2e^-$ ORR to H_2O_2 proceeds at 0.70 V vs. RHE, generating an oxidant that subsequently initiates MOR via Fenton-derived •OH radicals (an exothermic process). On the anodic side, GOR occurs at a substantially lower potential of approximately 0.1–0.4 V vs. RHE, depending on the oxygenated products formed [27, 28]. In particular, the formation of formic acid proceeds at E = 0.14 V vs. RHE, which is far less demanding than the OER (E = 1.23 V vs. RHE) [21, 29]. The favorable potential difference between the cathodic $2e^-$ ORR and the anodic GOR establishes a positive cell potential, indicating that the MOR-GOR full-cell system can operate spontaneously. Incorporating a photoanode further enhances this process by supplying photovoltage to overcome kinetic overpotentials, thereby enabling operation without the need for an external bias. By coupling these two complementary half-reactions, the design minimizes energy requirements, circumvents the high overpotential of OER, and simultaneously enables the co-production of value-added liquid products at both electrodes.

2.2 | Methane Oxidation Via In situ Electron-Fenton Chemistry

Oxidized carbon black (o-CB) was selected as the cathode catalyst, owing to its high surface area and ability to promote the $2e^-$ ORR toward H_2O_2 [30–32]. The catalyst was prepared by refluxing commercial carbon black in concentrated nitric acid, introducing oxygen-containing functional groups that enhance H_2O_2 selectivity [33, 34]. X-ray photoelectron spectroscopy (XPS) confirmed the formation of ethers (C—O—C), hydroxyls (C—OH), and carboxylic acids (HO—C═O) on the catalyst surface, while transmission electron microscopy (TEM) revealed no major change in nanoparticle morphology (Figures S1 and S2). Linear sweep voltammetry (LSV) in O_2-saturated 0.1 M Na_2SO_4 electrolyte (adjusted to pH 2 using 0.5 M H_2SO_4) showed that o-CB-coated carbon cloth exhibited significantly higher current densities than bare carbon cloth (Figure 2a). Faradaic efficiencies (FE) for H_2O_2 exceeded 80% across a wide potential range (Figure 2b), confirming its suitability for selective H_2O_2 production in acidic media. It should be noted that the H_2O_2 FE tends to decrease as the product accumulates and is further reduced to H_2O, but this secondary reduction is not a significant concern for our design, as the H_2O_2 is converted in situ for the immediate formation of the •OH radical through the Fenton reaction (Figure S3).

To drive CH_4 oxidation using in situ generated H_2O_2, Fe^{2+} was added to catalyze the Fenton reaction for •OH formation. Since Fe^{2+} cannot be further reduced under the reaction conditions, negligible change in the current density was observed upon Fe^{2+} addition (Figure S4). Without Fe^{2+}, negligible liquid products were detected, indicating that H_2O_2 alone was insufficient to

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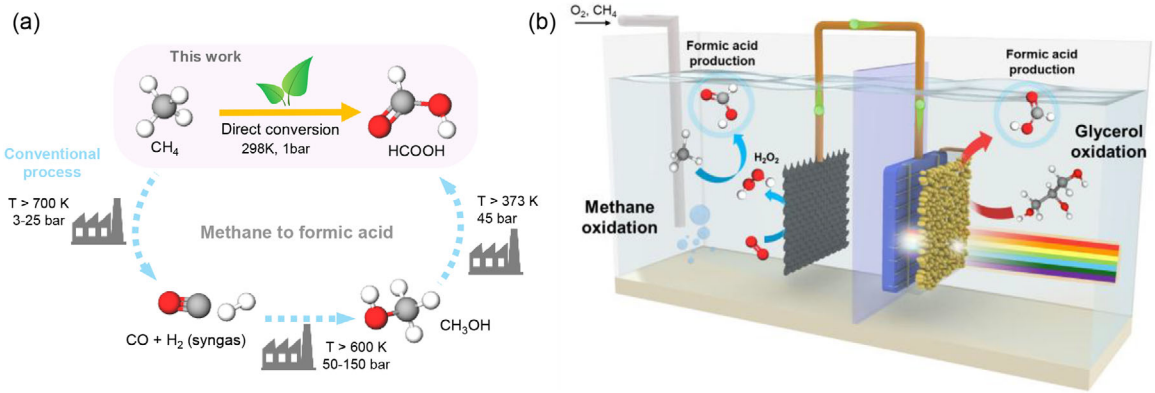


FIGURE 1 | (a) Conventional multi-step syngas route and the proposed direct process for converting CH_4 to HCOOH. (b) Schematic of a bias-free PV-PEC tandem system coupling glycerol oxidation reaction (GOR) with methane oxidation reaction (MOR).

TABLE 1 | Theoretical $\Delta G°$ and $E°_{cell}$ values of the cathodic $2e^-$ ORR for cascade MOR, coupled to anodic OER or GOR.

Cathode Reactions	Possible Anode Reactions	$\Delta G°_{reaction}$ (kJ/mol)	$E°_{cell}$ [V]
$2e^-$ ORR ($E° = 0.7$ V vs. RHE) $O_2 + 2H^+ + 2e^- \rightarrow H_2O_2$ **Fenton-assisted MOR** (Spontaneous) $H_2O_2 + Fe^{2+} + H^+ \rightarrow \bullet OH + Fe^{3+} + H_2O$ $CH_4 + 6\bullet OH \rightarrow HCOOH + 4H_2O$	**OER** ($E° = 1.23$ V vs RHE) $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$	206.2	−0.53
	GOR ($E° = 0.14$ V vs RHE) $C_3H_8O_3 + 3H_2O \rightarrow 3HCOOH + 8H^+ + 8e^-$	−432.0	0.56

Note: the thermodynamic properties of the reactions were calculated from the species data reported in Tables S1 under standard conditions (1 bar and 298 K). $\Delta G°_{reaction}$ calculations only included the electrode reactions, not considering the MOR reaction in the electrolyte. The MOR reaction shown represents the overall stoichiometric oxidation pathway to HCOOH. The actual process proceeds through sequential radical-mediated elementary reactions.

activate CH_4. In the absence of H_2O_2 generation, negligible CH_4 oxidation products were detected, indicating that H_2O_2 formation, followed by •OH generation via the electron-Fenton reaction, is essential for CH_4 activation (Figure S5). Upon adding Fe^{2+}, liquid products including methanol (CH_3OH), methyl hydroperoxide (CH_3OOH), and HCOOH were observed (Figure 2c), while no detectable formaldehyde was identified under the present reaction conditions (Figure S6). Notably, 0.2 mM Fe^{2+} achieved the highest liquid product selectivity of 84.8% and minimal CO_2 formation. HCOOH was the dominant product, reaching a production rate of 3.45 mmol g_{cat}^{-1} h^{-1}, surpassing most reported CH_4 oxidation systems under ambient conditions (Figure S7). In contrast, 1 mM Fe^{2+} showed a high liquid production rate but also resulted in increased CO_2 production, leading to a lower liquid selectivity of 56.7%. This is attributed to the increased Fenton reaction rate, which generates higher concentrations of •OH, thereby favoring further CH_4 oxidation and over-oxidation to CO_2. Increasing Fe^{2+} above 1 mM reduced total product yields, likely due to (i) non-productive radical scavenging by excess Fe^{2+} ($Fe^{2+} + \bullet OH \rightarrow Fe^{3+} + OH^-$) and (ii) competitive Fe^{3+} reduction at the cathode that diverts electrons from the $2e^-$ ORR (Figure S8).

Applied potential strongly influenced product distribution (Figure 2d). At more negative potentials, total product yield increased, but liquid selectivity dropped, as excess •OH can promote overoxidation to CO_2. Increasing the electrolyte volume suppressed CO_2 formation by diluting radical concentration and prolonging liquid product stability (Figure S9). Adjusting the CH_4/O_2 feed ratio revealed that a 1:1 flow provided the optimal balance between H_2O_2 generation and CH_4 availability, maximizing overall product yields (Figure S10).

Mechanistic studies confirmed the unique role of Fe^{2+}. Among tested transition metal ions (Cu^{2+}, Co^{2+}, Ni^{2+}, Mn^{2+}, Zn^{2+}, and Ce^{3+}), only Fe^{2+} enabled liquid product formation (Figure 3a). Ex situ electron paramagnetic resonance (EPR) spectroscopy using 5,5-Dimethyl-1-pyrroline N-oxide (DMPO) as a spin trap detected the characteristic quartet of DMPO-OH adducts exclusively in Fe^{2+}-containing systems, with no such signal observed for other metal ions (Figure 3b and Figure S11). In situ EPR during electrolysis further confirmed continuous •OH generation under applied bias in the presence of Fe^{2+} (Figure 3c). Furthermore, the addition of a radical scavenger completely suppressed the formation of oxygenated liquid products, further supporting that Fe^{2+}-mediated Fenton chemistry and the associated reactive oxygen species provide an operative pathway for CH_4 activation (Figure S12). Moreover, isotopic labeling experiments using $^{13}CH_4$ clearly demonstrate that the produced HCOOH originated directly from CH_4 oxidation (Figure S13).

Control experiments with externally added H_2O_2 showed that o-CB alone could not activate H_2O_2 to •OH under cathodic bias (Figure 3d; see Experimental Section for details). When Fe^{2+} was present, CH_4 oxidation was initiated via the Fenton reaction with •OH generation, and the product yields further

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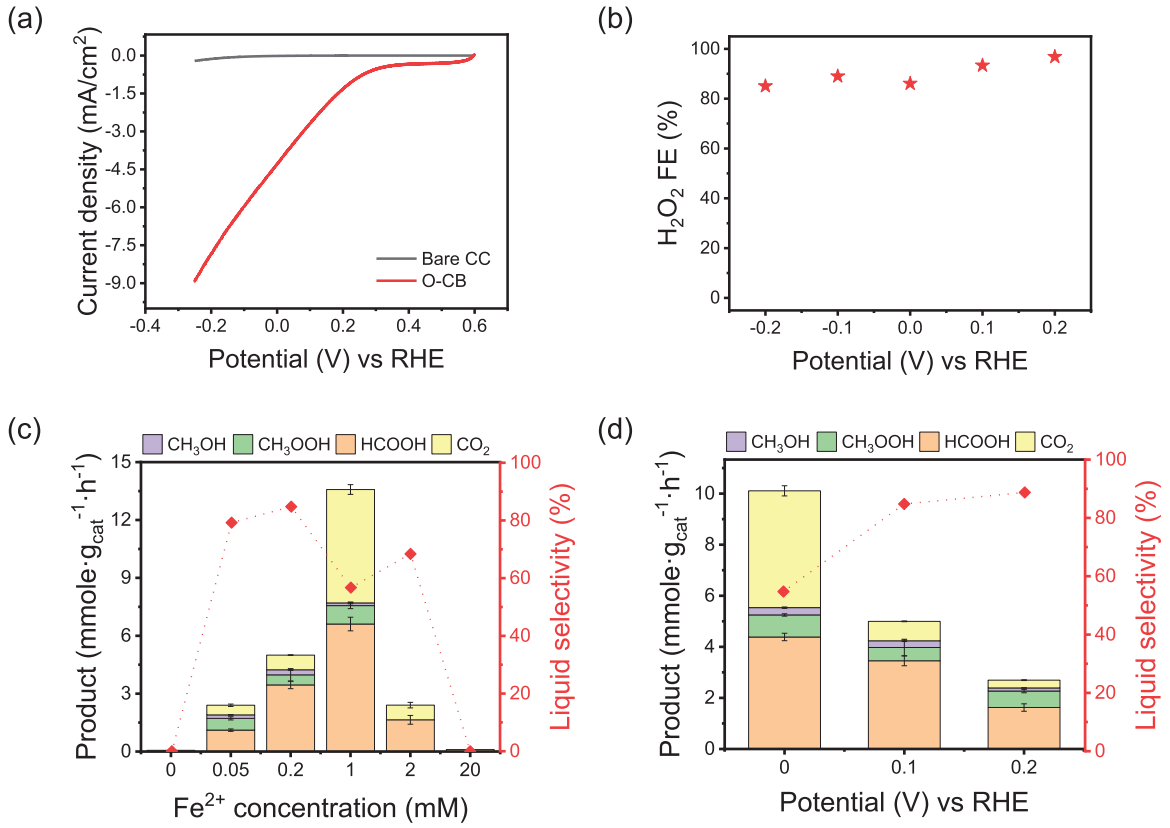


FIGURE 2 | (a) LSV curves of bare carbon cloth and o-CB coated carbon cloth in O_2-saturated 0.1 M Na_2SO_4 (pH = 2) electrolyte (scan rate: 10 mV/s). (b) Faradaic efficiency for H_2O_2 production during ORR on o-CB at various applied potentials. (c) CH_4 oxidation product yield with varying Fe^{2+} concentrations (applied potential = 0.1 V vs. RHE). (d) CH_4 oxidation product yield under different applied potentials (Fe^{2+} = 0.2 mM).

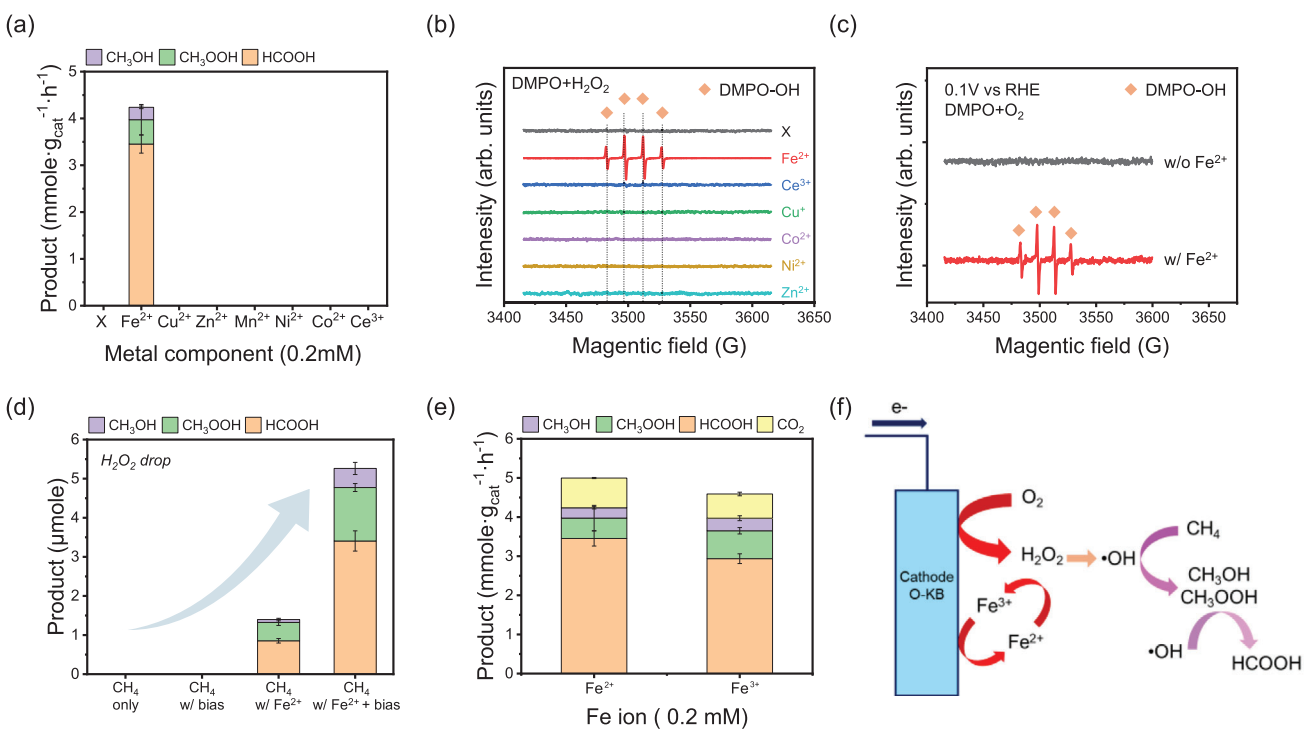


FIGURE 3 | (a) Comparison of CH_4 oxidation product yield with 0.2 mM of various metal ions. (b) Ex situ EPR spectra of various 5 mM metal ions reacting with 10 mM H_2O_2 in the presence of DMPO. (c) In situ EPR spectra of ORR with and without Fe^{2+}. (d) H_2O_2 drop-in control experiment with CH_4 flow under various conditions. (e) Comparison of CH_4 oxidation product yield using Fe^{2+} and Fe^{3+}. (f) Proposed reaction pathway for CH_4 oxidation via in-situ electron-Fenton chemistry.

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increased under applied bias. This suggests that Fe^{3+}, formed during the Fenton reaction, is electrochemically reduced back to Fe^{2+}, thereby sustaining the catalytic cycle and promoting continuous •OH generation. Similar product yields were obtained when Fe^{3+} was added instead of Fe^{2+}, supporting the regeneration mechanism (Figure 3e). It should be noted that the Fe^{3+} reduction current is a minor contribution compared to the ORR current, thereby not affecting the production rate of H_2O_2 (Figure S14). Moreover, the control experiment using a non-acidic electrolyte showed negligible CH_4 oxidation with externally added H_2O_2. This confirms that the Fenton reaction is only effective in an acidic environment (Figure S15).

Based on these results, we propose the mechanism illustrated in Figure 3f. O_2 is first electrochemically reduced at the o-CB cathode via the $2e^-$ ORR to H_2O_2. In the presence of Fe^{2+}, H_2O_2 undergoes the Fenton reaction to produce •OH radicals, which abstracts hydrogen from CH_4 to form CH_3• radicals. These CH_3• radicals can further react with dissolved O_2 and reactive oxygen species (•OH and •OOH), leading to the formation of CH_3OOH, CH_3OH, which are subsequently oxidized to HCOOH [35]. Control experiments further confirmed that CH_3OH can be sequentially oxidized to HCOOH under the reaction conditions (Figure S16). Notably, CH_3OH is more readily oxidized by •OH than HCOOH due to its larger reaction rate constant, while HCOOH exhibits comparatively slower further oxidation and enhanced stabilization due to solvation effects [36, 37]. Thus, HCOOH preferentially accumulates as the dominant liquid product in the cascade process.

2.3 | Photoanodic Glycerol Oxidation

To replace the commonly used energy-intensive OER and enable bias-free CH_4 oxidation, we selected glycerol oxidation as the anodic reaction, due to its lower thermodynamic barrier and potential to generate value-added products [20, 23]. Among tested photoanodes, nanoporous $BiVO_4$ exhibited the largest negative onset potential shift and the greatest photocurrent enhancement upon glycerol addition, suggesting efficient utilization of photo-generated holes for organic oxidation rather than water oxidation (Figure 4a). $BiVO_4$ also demonstrated the highest catalytic activity for glycerol oxidation in the dark (Figure S17). Material characterization confirmed its morphological, optical, and crystalline features suitable for PEC operation (Figure S18).

When tested in an H-type cell with 0.1 M Na_2SO_4 electrolyte (pH 2), $BiVO_4$ maintained stable photocurrents for 3 h under simulated sunlight (AM 1.5G, 1 sun) in the presence of glycerol (Figure 4b). Extended stability data, showing stable performance for over 15 h, are provided in Figure S19. This stability contrasts sharply with the rapid degradation observed during water oxidation, as glycerol efficiently scavenged photo-generated holes and thereby suppressed photocorrosion [38]. Electrochemical impedance spectroscopy further confirmed faster interfacial hole-transfer kinetics in glycerol-containing electrolyte (Figure 4c), which would facilitate sustained current flow in the integrated full-cell configuration.

Product analysis by HPLC identified multiple partially oxidized products, including glycolic acid (GOA), HCOOH, glyceric acid (GEA), dihydroxyacetone (DHA), glycoaldehyde (GCAD), and glyceraldehyde (GLAD), with an overall FE approaching 100% and CO_2 accounting for < 2% (Figure 4d and Figure S20). Notably, the C_1 (HCOOH) to C_2 (GCAD and GOA) product ratio remained constant (~1: 2) across tested potentials, suggesting concurrent C—C bond cleavage and C—C coupling reaction pathways (Figure S21) [29, 39]. This indicates that product selectivity is largely independent of applied bias, simplifying process integration with the cathodic half-reaction.

2.4 | Bias-Free Operation of MOR-GOR Tandem Cell

To integrate the two half-reactions into an unbiased full-cell system, we constructed a PV-PEC tandem device by coupling the $BiVO_4$ photoanode with an underlying Si solar cell (Figure S22 and Table S2), providing sufficient photovoltage (~0.58 V) to boost cathodic H_2O_2 formation without external bias (Figure 5a and Figure S18f). The two electrodes were separated by an anion exchange membrane (FAB-PK-130) to prevent glycerol crossover while allowing ion conductivity (Figure S23). It should be noted that while a slight permeation of glycerol through the anion exchange membrane may occur, the amount is negligible (Figure S24). On the cathode side, a larger catholyte volume was used to accumulate CH_4 oxidation products and suppress overoxidation (Figure S25). Under AM 1.5G illumination, the tandem device achieved bias-free operation at a current density of 1.7 mA cm^{-2} (Figure 5b), and remained stable over 15 h of continuous operation in five consecutive 3 h cycles (Figure 5c). While continuous bias-free operation was maintained during the 15 h test, a slight dissolution of the $BiVO_4$ photoanode was observed after operation (Figure S26), which contributed to the upward trend in the stability test (Figure 4b and Figure S19). However, this issue may be further mitigated through electrolyte engineering or surface passivation, whereas the o-CB cathode showed negligible morphological or compositional degradation (Figure S27) [40, 41]. In each cycle, the cathode side produced 2.87 mmol g_{cat}^{-1} h^{-1} HCOOH from MOR, while the anode generated additional HCOOH and other oxygenated products from GOR, resulting in a total of 7.99 μmol cm^{-2} h^{-1} HCOOH (Figure 5c,d and Figure S28 and Table S3). The desirable conversion of GOR to FA is attributed to the strong adsorption of glycerol and its oxygenates, and this selectivity can be further enhanced by increasing this preferential adsorption [42, 43]. Notably, the findings highlight the potential for further optimization of electrolyte and photoanode design may enable bias-free operation without a solar cell by reducing (photo)electrochemical overpotential (Figure S29).

2.5 | Life Cycle Assessment

LCA compared four routes for HCOOH production from methane: (1) conventional syngas process (CH_4 → syngas → CH_3OH → HCOOH); (2) H_2O_2-mediated MOR paired with OER in an electrochemical cell (MOR-OER); (3) H_2O_2-mediated MOR paired with GOR in an electrochemical cell; and (4) the present MOR-GOR in a PV-PEC system. Using the average U.S. electricity grid mix [44], the syngas route emitted 2.1 kg CO_2e per kg HCOOH, the MOR-OER (EC) and MOR-GOR (EC) processes emitted 24.5 and 12.7 kg CO_2e per kg HCOOH due to high

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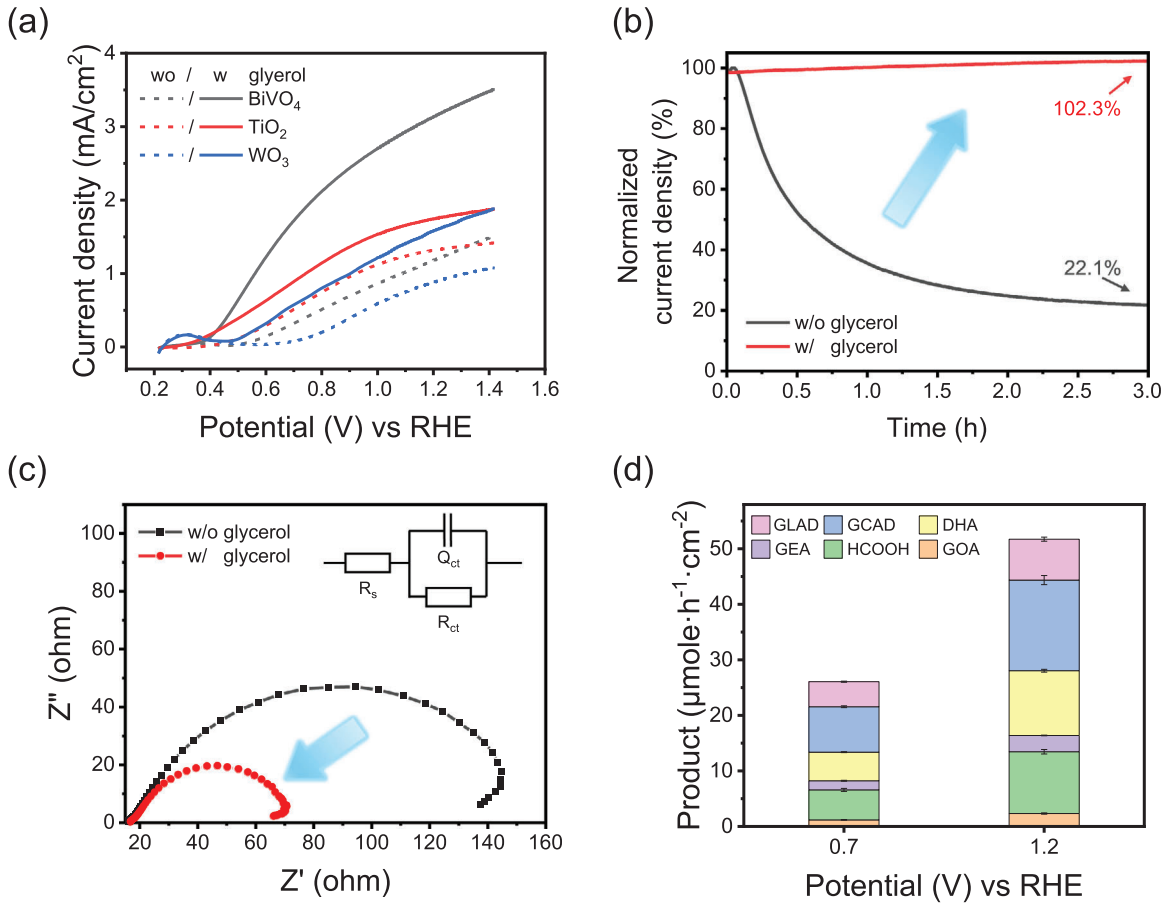


FIGURE 4 | (a) LSV curves of various photoanodes with and without 0.1 M glycerol in 0.1 M Na_2SO_4 (pH = 2) under illumination (scan rate: 25 mV/s). (b) Normalized current density-time graph for the $BiVO_4$ photoanode with and without 0.1 M glycerol at 0.7 V vs RHE. The normalized current density was calculated as the percentage of the initial current density ($j_t / j_{initial}$ x 100%). (c) EIS Nyquist plot of the $BiVO_4$ photoanode at 0.7 V vs. RHE in with and without 0.1 M glycerol. (d) Comparison of glycerol oxidation product yield after 3 h of PEC conversion at applied potentials of 0.7 V and 1.2 V vs RHE.

electricity consumption, and the MOR-GOR PV-PEC system emitted only 1.1 kg CO_2e per kg HCOOH (Figure 5e). When powered entirely by solar electricity, total CO_2 emissions dropped to 1.6, 3.6, 2.7, and 0.5 kg CO_2e per kg HCOOH for the syngas, MOR-OER (EC), MOR-GOR (EC), and MOR-GOR (PV-PEC) routes, respectively. Excluding indirect emissions from electricity and heating utilities, the MOR-GOR (PV-PEC) reaction process itself produced 0.4 kg CO_2e per kg HCOOH, corresponding to a reduction of over 66% compared to 1.5 kg CO_2e per kg HCOOH from the syngas route (Figure 5f). Further information such as LCA parameters and mass balances can be found in Tables S4–S10.

3 | Conclusions

We have demonstrated the first unbiased PV-PEC tandem system for simultaneous methane and glycerol upgrading to HCOOH under ambient conditions. Coupling in situ electro-Fenton methane oxidation at the o-CB cathode with photoanodic glycerol oxidation at $BiVO_4$ delivers a high CH_4-to-HCOOH conversion rate of 2.87 mmol g_{cat}^{-1} h^{-1} and a total HCOOH production rate of 7.99 μmol cm^{-2} h^{-1} for the full cell, which showed stable operation over 15 h and more than 66% reduction in greenhouse gas emissions as confirmed by LCA. These results establish a versatile and sustainable platform for bias-free, solar-driven co-upgrading of underutilized carbon resources, paving the way toward decentralized methane and biomass utilization.

4 | Experimental Section

4.1 | Preparation of o-CB Cathode

Conductive carbon black (EC600JD, MSE Supplies, 500 mg) was dispersed in 250 mL of 12 M HNO_3 (70%, Fisher chemical) and maintained at 80°C in a heating mantle under continuous stirring for 24 h. After the oxidative treatment, the oxidized carbon black (o-CB) was collected by filtration, thoroughly rinsed with deionized water and ethanol, and dried overnight in a convection oven.

Catalyst ink was prepared by suspending 4 mg of o-CB in a solvent mixture of 0.2 mL of water and 0.8 mL of ethanol containing 40 μL of 5 wt.% Nafion solution (Sigma–Aldrich). The mixture was sonicated until a homogeneous dispersion was obtained. The ink was drop-cast onto carbon cloth (3 cm^2, Fuel Cell Store) and dried under ambient conditions. The mass loading of o-CB was

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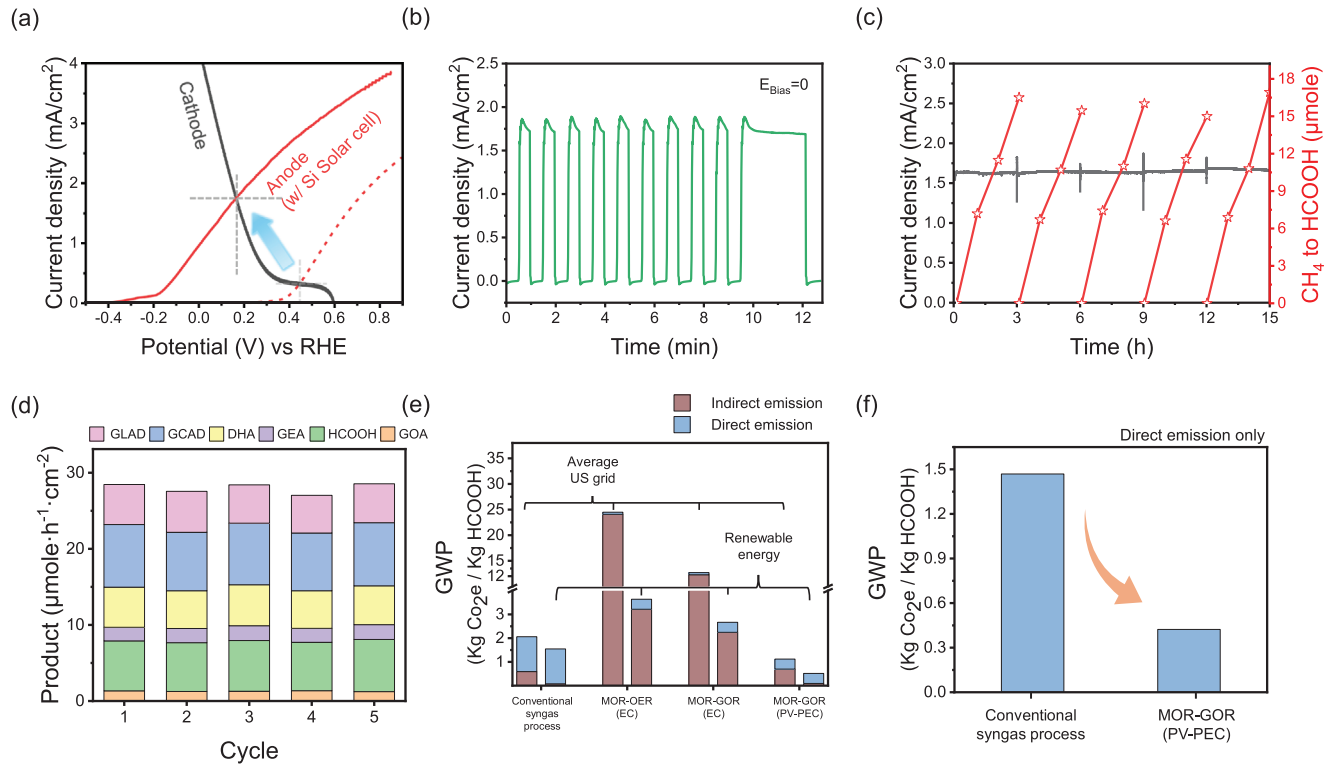


FIGURE 5 | (a) Coupled LSV curves of CH_4 oxidation at the o-CB cathode and glycerol oxidation at the $BiVO_4$ photoanode, with and without integration of Si solar cells. (b) Chronoamperometric response of the PV-PEC system under bias-free conditions. (c) Cycling test of the bias-free system for simultaneous CH_4 and glycerol oxidation, showing current density (black) and CH_4 to HCOOH production yield (red) over time. (d) Photoanodic glycerol oxidation yield after each cycle of the bias-free system. (e) Global warming potential (GWP) comparison for CH_4 to HCOOH production via the conventional syngas process route, the MOR-OER paired electrochemical (EC) cell, the MOR-GOR paired EC cell, and the MOR-GOR paired PV-PEC cell. (f) GWP comparison for the same processes, showing direct CO_2 emissions only.

0.5 mg cm^{-2}, determined by weighing the electrode before and after deposition.

4.2 | Synthesis of Photoanodes

$BiVO_4$ photoanodes were synthesized via a two-step process with minor modifications to a previously reported method [41]. Fluorine-doped tin oxide (FTO) substrates (TEC 7, MSE Supplies) were cleaned by sequential sonication in acetone, isopropanol, and deionized water for 10 min each. The BiOI precursor solution was prepared by mixing 0.4 M KI (99%, Sigma–Aldrich) and 0.04 M $Bi(NO_3)_3$ $5H_2O$ (98%, Sigma–Aldrich) in deionized water, followed by the addition of HNO_3 to adjust the pH to 1.75. This solution was then mixed with absolute ethanol containing 0.23 M p-benzoquinone (98%, Sigma–Aldrich) and stirred vigorously for several minutes. Electrodeposition of BiOI was carried out in a standard three-electrode cell at −0.1 V vs. Ag/AgCl for 200 s. Subsequently, a DMSO solution containing 0.2 M $VO(acac)_2$ (98%, Sigma–Aldrich) was drop-cast onto the BiOI-coated FTO substrate, followed by thermal treatment in air at 450°C for 2 h to yield $BiVO_4$. The resulting films were immersed in 1 M NaOH and gently stirred for 30 min to remove excess V_2O_5. TiO_2 and WO_3 photoanodes were synthesized via hydrothermal growth and flame vapor deposition, respectively, following previously reported procedures [45, 46].

4.3 | Methane Oxidation Reaction (MOR) Experiments

MOR experiments were performed in a custom-designed H-type cell separated by a Nafion 115 membrane at room temperature. A platinum coil served as the counter electrode, and an Ag/AgCl (saturated KCl) electrode was used as the reference. The electrolyte consisted of 50 mL of 0.1 M Na_2SO_4, with the pH adjusted to 2 using 0.5 M H_2SO_4. Reactions were conducted for 1 h at constant applied potentials with controlled CH_4 and O_2 flow rates. Fe^{2+} concentrations were varied (0–20 mM) as specified. For product accumulation, 400 mL of catholyte was continuously circulated at 15 mL min^{-1} during the reaction.

4.4 | H_2O_2 Drop-in Experiments

The H_2O_2 drop-in experiment was performed by adding 10 mL of 0.1 M Na_2SO_4 solution (pH 2) containing 10 μL of 30% H_2O_2 dropwise into 40 mL of Na_2SO_4 (pH 2) over 1 h under vigorous stirring and continuous CH_4 purging. The H_2O_2 concentration was matched to the estimated in situ generation based on electrochemical measurements and H_2O_2 Faradaic efficiency calculation. When specified, 0.2 mM Fe^{2+} and an applied potential of 0.1 V vs. RHE were introduced.

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4.5 | Bias-Free Tandem PV-PEC System

The bias-free operation was conducted in a two-electrode configuration using a two-compartment reactor under simulated AM 1.5G one sun illumination. The anode chamber contained 50 mL of 0.1 M Na_2SO_4 (pH = 2) with 0.1 M glycerol, while the cathode chamber contained 400 mL of 0.1 M Na_2SO_4 (pH 2), which was continuously circulated at 15 mL min^{-1}. The chambers were separated by an anion exchange membrane (FAB-PK-130, Fuel Cell Store), pre-activated by immersion in 0.5 M NaCl solution for 24 h.

The $BiVO_4$ photoanode was electrically connected to the Si solar cell via soldering, and the assembly was sealed using a Si gasket to prevent electrolyte intrusion. During operation, CH_4 and O_2 were continuously co-fed into the cathode chamber, each at a flow rate of 100 sccm. The active area of the photoanode, Si solar cell, and o-CB cathode was maintained at 3.5 cm^2. The cathode electrolyte was pre-saturated with CH_4 and O_2 for 30 min before operation.

4.6 | Product Analysis and Material Characterizations

Electrochemical measurements were performed using a potentiostat (Gamry 1010E). Gaseous products (CO_2 and CO) were quantitatively analyzed using gas chromatography (GC 8860, Agilent, Figure S30). Liquid products (HCOOH, CH_3OH and CH_3OOH) from MOR experiments were quantified using ^1H NMR (B600 Bruker NEO Spectrometer) with water suppression. Product assignment was confirmed by comparing NMR signals with high-purity commercial standards. The corresponding chemical shifts were further validated by spiking the reaction mixture with reference standards to ensure peak consistency. For NMR analysis, a 0.5 mL aliquot of the reaction electrolyte was mixed with 0.1 mL of D_2O (Cambridge Isotope Laboratories) and 25 µL of diluted dimethyl sulfoxide (Fisher Chemicals) as the internal standard. For the $BiVO_4$ photoanode during GOR, the results were consistent with literature values; therefore, replicate measurements and standard deviation calculations were not performed [29]. HPLC was utilized to analyze the products from glycerol oxidation using an ion exclusion column (Shimadzu Nexera, Bio-Rad Aminex HPX 87H) with 5 mM H_2SO_4 mobile phase flowing at 0.6 mL min^{-1}. The column temperature was maintained at 60°C, and separated products were detected using a UV–vis detector at a wavelength of 210 nm (GOA, FA, GEA) or 254 nm (DHA, GCAD), or a Refractive Index Detector (GLAD). Products were quantified using a MATLAB script and mathematics as described in prior work [24, 26, 47]. All reported potentials were converted to the reversible hydrogen electrode (RHE) scale using the equation: E (V vs. RHE) = E (V vs. Ag/AgCl) + (0.059 × pH) + 0.197 V. HCHO and H_2O_2 concentrations were quantified using colorimetric methods (Figures S6–S31) [48, 49]. Tert-butyl alcohol (10 mM) and p-benzoquinone (10 mM) were employed to capture •OH and •OOH, respectively. The crystal structures of photoanodes were characterized by XRD (Bruker D8 Advance). SEM (Hitachi S-4800) was used to obtain the morphology of electrodes. Tauc plots were derived by UV–vis spectrometry (Agilent Cary 5G). XPS (Kratos Axis ULTRA) was performed to analyze the surface components of o-CB and photoanodes. The acquired XPS spectra were processed in CasaXPS software using Shirley background subtraction and Lorentzian-Gaussian peak fitting. Electrochemical impedance spectroscopy measurements were performed at 0.7 V vs. RHE in 0.1 M Na_2SO_4 (pH 2) by applying a 10 mV AC amplitude over a frequency range of 100 kHz to 100 mHz. Both in situ and ex situ EPR measurements (Bruker, EMXPlus X-band Continuous Wave) were conducted using high-purity quartz capillary tubes. For in situ measurements, radicals generated during the electro-Fenton reaction were captured with a Fe^{2+} concentration of 0.2 mM, consistent with the MOR experimental conditions. For ex situ measurements, H_2O_2 was added as a reference oxidant together with various ions at the concentration of 5 mM to facilitate identification of the radical species. In both experiments, 100 mM DMPO (Fisher Scientific) was employed as the spin-trap agent. All reported yields and selectivity results were obtained from at least three independent measurements, and the data are presented as mean values with error bars representing the standard deviation.

$$\text{Liquid product selectivity} = 100\ (\%) * \frac{n(CH_3OOH) + n(CH_3OH) + n(HCOOH)}{n(CH_3OOH) + n(CH_3OH) + n(HCOOH) + n(CO_2)}$$

4.7 | Life Cycle Assessment

The scope of the LCA is defined as a cradle-to-gate system, starting from natural gas extraction and ending with formic acid production. Recycling pathways for byproducts from methanol and formic acid synthesis were not considered in this study. The primary goal of the LCA is to compare CO_2 emissions among three routes: the conventional formic acid process, the MOR–OER process, and the MOR–GOR process. While other LCA impact categories (acidification, eutrophication, ecotoxicity, resource depletion, land use, water use, and human health effects) are also important, this modeling focused exclusively on CO_2-equivalent emissions. The functional unit was defined as the normalized production of 1 kg of formic acid. Due to the lack of liquid separation data for GOR products, only the one-cell operation of the MOR process was considered in this study. If the GOR reactor is analyzed in future studies, additional factors such as the global warming potential (GWP) associated with glycerol input, CO_2 emissions from the GOR process, and the emissions from the liquid product separation process should be considered for a comprehensive assessment.

In the conventional formic acid process, natural gas (~95% CH_4) is extracted and converted via steam reforming, where O_2 from air separation and steam are used to produce synthesis gas (syngas). The syngas is subsequently processed in CH_3OH reactor, followed by distillation to obtain pure CH_3OH. For HCOOH production, pure CH_3OH and CO are utilized, with approximately 70 wt.% of the CH_3OH being recycled back into the process to sustain continuous HCOOH synthesis (Figure S32). The input and output data for the conventional CH_3OH production process were obtained from previously published LCA studies [50, 51]. The LCA for the conventional HCOOH production process was modeled based on the previous framework [52]. The LCA results were calculated using normalized input data

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from CH_3OH production to represent the upstream inventory for HCOOH synthesis. Additionally, the amount of make-up CH_3OH required in the process was based on reported data [53]. To evaluate both direct and indirect emissions, CO_2 emissions were categorized into those derived from electricity and heat, and those excluding these energy sources. Indirect emissions were calculated using U.S. average grid electricity emission factors [44].

Output data for the MOR-OER and MOR-GOR processes were collected from experimental results. All LCA data for the bias-free MOR were determined from one-hour electrochemical experiments conducted with a static 50 mL cathodic electrolyte volume, focusing specifically on gaseous product quantification. For the LCA calculation, unreacted CH_4 was assumed to be recycled and subsequently converted with the same product selectivity, assuming complete overall CH_4 utilization. Similar to the conventional pathway, energy consumption associated with natural gas extraction was included [54]. Although pure O_2 was used in the lab-scale experiments, the energy requirement for air separation was considered for realistic application scenarios. Furthermore, energy consumption for HCOOH separation heat was included, since the liquid products consist of a mixture of compounds that require purification [55]. For direct emissions, CO_2 generation was normalized by the amount of HCOOH produced. Indirect emissions were calculated using the same methodology applied to the conventional HCOOH production process. To focus specifically on the CH_4-to-HCOOH conversion pathway, CO_2 emissions associated with glycerol oxidation were omitted from the comparison presented in Figure 5e,f. However, the complete LCA results, which incorporate the CO_2 footprint associated with glycerol utilization and direct reaction emissions normalized to HCOOH production, are provided in Table S10.

Author Contributions

Ho Kun Woo: conceptualization, methodology, data curation, validation, investigation, Writing – original draft, Writing – review and editing. **Beomhui Lee**: data curation, writing – review and editing, formal analysis. **Rachel N. Gaines**: data curation, writing – review and editing, formal analysis. **Songtao Tang**: writing – review and editing. **Lili Cai**: writing – review and editing, writing – original draft, funding acquisition, project administration, resources, supervision, conceptualization.

Acknowledgements

The research was financially supported by the donors of the ACS Petroleum Research Fund (PRF) under Doctoral New Investigator Grant 65618-DNI5. L.C. served as Principal Investigator on ACS PRF 65618-DNI5 that provided support for H.K.W. P.J.A.K. served as Principal Investigator on NSF EFRI-DCheM 2029326 and co-Principal Investigator on NSF FMRG-Cyber 2328160 that both provided support for R.N.G. This research was also financially supported by the Center for Advanced Climate Solutions (CACS) at the University of Illinois Urbana-Champaign. TEM, XRD, SEM, XPS, and UV–vis were carried out in Materials Research Laboratory at the University of Illinois. H-NMR was performed in the School of Chemical Sciences NMR Lab at the University of Illinois. Major funding for the Bruker EMXPlus was provided by National Science foundation award 1726244 (2017) to the School of Chemical Sciences EPR lab at the University of Illinois.

Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

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